

Fluid Kinematics - Question Bank

Unit III - Mechanical Engineering Department

2 MARKS QUESTIONS

- Q1. Define fluid and state any two characteristics of fluid.
- Q2. What is viscosity? Write Newton's law of viscosity.
- Q3. Define density and specific gravity.
- Q4. Distinguish between cohesion and adhesion in liquids.
- Q5. What is surface tension? How is it related to cohesion?
- Q6. Define capillarity.
- Q7. What is compressibility of a fluid?
- Q8. Distinguish between Newtonian and Non-Newtonian fluids.
- Q9. Give two examples each of Newtonian and Non-Newtonian fluids.
- Q10. What is fluid kinematics?
- Q11. Define Lagrangian method of fluid flow analysis.
- Q12. Define Eulerian method of fluid flow analysis.
- Q13. What is a path line in fluid mechanics?
- Q14. What is a field variable? Give any two examples.
- Q15. Define steady flow.
- Q16. Define unsteady flow.
- Q17. Distinguish between steady and unsteady flow.
- Q18. Define uniform flow.
- Q19. Define non-uniform flow.
- Q20. Distinguish between uniform and non-uniform flow.
- Q21. What is laminar flow? Give one example.
- Q22. What is turbulent flow? Give one example.

- Q23. Distinguish between laminar and turbulent flow.
- Q24. Define compressible flow with an example.
- Q25. Define incompressible flow.
- Q26. What is discharge in fluid mechanics? Write its unit.
- Q27. Write the continuity equation for incompressible flow.
- Q28. State the continuity equation for steady flow.
- Q29. Write the unit of viscosity in SI system.
- Q30. What is rate of shear deformation?
- Q31. Define velocity gradient.
- Q32. Write the expression for mass flow rate in a pipe.
- Q33. What condition makes $\frac{dm}{dt} = 0$ in fluid flow?
- Q34. Write the general form of continuity equation: $\partial(\rho AV) = ?$
- Q35. In which coordinate system are fluid particles tracked individually?
- Q36. What is the main advantage of Eulerian method over Lagrangian method?
- Q37. If velocity field is $V = (3x + 2y)i + (2z + 3x^2)j + (2t - 3z)k$, what is the component u ?
- Q38. For incompressible steady flow, if $A_1V_1 = A_2V_2$, what remains constant?
- Q39. What is rheology?
- Q40. State any two properties of liquids important in hydraulics.

3 MARKS QUESTIONS

- Q1. Explain Newton's law of viscosity with a neat diagram.
- Q2. Differentiate between Newtonian and Non-Newtonian fluids with examples.
- Q3. Explain the Lagrangian method of describing fluid motion.
- Q4. Explain the Eulerian method of describing fluid motion with the concept of control volume.
- Q5. Compare Lagrangian and Eulerian methods of fluid flow analysis.
- Q6. What is a field variable? Explain how velocity is expressed as a field variable in Eulerian method.
- Q7. Distinguish between steady and unsteady flow with suitable examples.
- Q8. Distinguish between uniform and non-uniform flow with suitable examples.
- Q9. Explain laminar flow and turbulent flow with real-life examples.
- Q10. Differentiate between compressible and incompressible flow with examples.
- Q11. Define discharge and derive the relation $Q = AV$.
- Q12. Derive the continuity equation for one-dimensional steady flow.
- Q13. State and explain the principle of conservation of mass in fluid flow.
- Q14. Explain why $\rho_1 A_1 V_1 = \rho_2 A_2 V_2$ for steady flow.
- Q15. For incompressible flow, prove that $A_1 V_1 = A_2 V_2$.
- Q16. List and explain any three important properties of liquids.
- Q17. What is viscosity? Explain the concept of shear stress between fluid layers.
- Q18. Explain the term velocity gradient with a diagram showing two fluid layers.
- Q19. Why is Lagrangian method considered time-consuming for fluid flow analysis?
- Q20. In a velocity field $V = (3x + 2y)i + (2z + 3x^2)j + (2t - 3z)k$, identify whether the flow is steady or unsteady. Justify your answer.
- Q21. Explain the physical significance of the continuity equation.
- Q22. What happens to velocity when area of cross-section decreases in an incompressible flow? Explain.
- Q23. Define the three states of matter and explain what makes fluid different from solid.
- Q24. Explain cohesion, adhesion, and capillarity with their interrelation.
- Q25. Write the velocity components u , v , w in Eulerian method and explain their significance.

7 MARKS QUESTIONS

- Q1.** (a) Define fluid and list its characteristics.
(b) Explain any four important properties of liquids in detail.
- Q2.** (a) State Newton's law of viscosity.
(b) Differentiate between Newtonian and Non-Newtonian fluids with suitable examples.
(c) Draw and explain the graph showing relationship between shear stress and rate of deformation for different types of fluids.
- Q3.** (a) What is fluid kinematics?
(b) Explain Lagrangian and Eulerian methods of fluid flow description in detail.
(c) Compare the two methods and state which is more practical for fluid flow analysis.
- Q4.** (a) Define the following terms with examples:
(i) Steady and unsteady flow
(ii) Uniform and non-uniform flow
(iii) Laminar and turbulent flow
(iv) Compressible and incompressible flow
- Q5.** Derive the continuity equation for one-dimensional fluid flow. State the assumptions made and explain the physical significance of the equation.
- Q6.** (a) Define discharge in fluid mechanics.
(b) Derive the continuity equation starting from the principle of conservation of mass.
(c) Show that for incompressible steady flow, $A_1V_1 = A_2V_2$.
- Q7.** In a fluid, the velocity field is given by:
 $V = (3x + 2y)i + (2z + 3x^2)j + (2t - 3z)k$
Determine:
(i) The velocity components u , v , w at any point
(ii) The speed at point $(1, 1, 1)$
(iii) The speed at time $t = 2s$ at point $(0, 0, 2)$
(iv) Classify the velocity field as steady or unsteady, uniform or non-uniform, and one, two, or three dimensional.
- Q8.** The diameters of a pipe at sections 1-1 and 2-2 are 200 mm and 300 mm respectively. If the velocity of water flowing through the pipe at section 1-1 is 4 m/s, find:
(i) Discharge through the pipe
(ii) Velocity of water at section 2-2
Also explain the principle used in solving this problem.
- Q9.** (a) Explain the Eulerian method of fluid flow description.
(b) Write the velocity field equations in component form.
(c) What are the advantages of Eulerian method over Lagrangian method in practical fluid flow analysis?

- Q10.** (a) Define viscosity and state Newton's law of viscosity.
 (b) Explain the following properties of liquids:
 (i) Cohesion
 (ii) Adhesion
 (iii) Surface tension
 (iv) Capillarity
 (c) How are these properties interrelated?
- Q11.** (a) What is meant by steady flow and unsteady flow?
 (b) Distinguish between uniform and non-uniform flow.
 (c) Give practical examples for each type of flow.
 (d) Can a flow be steady and non-uniform simultaneously? Explain with an example.
- Q12.** Derive the general continuity equation $\partial(\rho AV) = 0$ from first principles. Starting from the mass balance between two sections of a pipe, show all the mathematical steps and state the final forms for:
 (i) Steady flow
 (ii) Incompressible flow
- Q13.** (a) Explain the concept of field variables in Eulerian method.
 (b) Write the general form of velocity as a function of space and time.
 (c) Differentiate between path line in Lagrangian method and field point in Eulerian method.
 (d) Why is Eulerian method preferred in most fluid mechanics problems?
- Q14.** (a) Define and explain laminar and turbulent flow with neat diagrams.
 (b) Give real-life examples of both types of flow.
 (c) What factors determine whether a flow will be laminar or turbulent?
- Q15.** (a) Explain the term 'rate of shear deformation' with a neat sketch.
 (b) State Newton's law of viscosity and derive the expression for shear stress.
 (c) Classify fluids based on their viscous behavior and plot the relationship between shear stress and velocity gradient for each type.